Chapter 5
Logic

The Mechanics of Proof | CSC302 lecture notes

Based on Heather Macbeth's textbook

What you should be able to do

- Check a propositional formula with a truth table and prove a logical equivalence.
- Track how quantifier order controls the dependence of witnesses.
- Use excluded middle without first deciding which case is true.
- Move negations through connectives and quantifiers correctly.
- Turn a negative statement into a concrete proof task.

Three levels of reasoning

Truth tables analyze propositions. Quantifiers introduce objects. Mathematics supplies facts about those objects.

Lean fragments use the textbook's educational tactic environment.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 5, Sections 5.1–5.3.

5.1: The connectives and their truth conditions

Let P and Q be propositions.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \to Q$	$P \leftrightarrow Q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

The one false implication case

$P \to Q$ fails exactly when P is true and Q is false. When P is false, there is no instance of P that violates the implication.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Example 5.1.2 and Exercise 5.1.3.

Build a compound truth table from the inside out

Consider $P \leftrightarrow (\neg P \vee Q)$.

P	Q	$\neg P$	$\neg P \vee Q$	$P \leftrightarrow (\neg P \vee Q)$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	F
F	F	T	T	F

Compute $\neg P$ first, then the disjunction, then the equivalence. The final column matches $P \wedge Q$.

Check versus explanation

For propositional formulas, matching truth tables establishes classical equivalence. A structural proof explains how to convert a proof in each direction.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Exercise 5.1.3 and Example 5.1.5.

Logical equivalence means mutual implication

Two statements A and B are logically equivalent when $A \leftrightarrow B$. For example, $(P \vee P) \leftrightarrow P$.

```
-- Demonstration: every branch of the disjunction gives P.
example (P : Prop) : Iff (Or P P) P := by
  constructor
  . intro h
    obtain h1 | h2 := h
    . exact h1
    . exact h2
  . intro h
    left
    exact h
```

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Example 5.1.4; ASCII teaching adaptation.

Distributing a conjunction over a disjunction

$$P \wedge (Q \vee R) \iff (P \wedge Q) \vee (P \wedge R).$$

Forward

We have P and either Q or R.

- If Q, build $P \wedge Q$ and choose the left alternative.
- If R, build $P \wedge R$ and choose the right alternative.

Backward

Split the given disjunction.

- From $P \wedge Q$, keep P and introduce $Q \vee R$ on the left.
- From $P \wedge R$, keep P and introduce $Q \vee R$ on the right.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Example 5.1.5.

A common witness works for every input

Let X, Y be types and let $P(x, y)$ be a proposition for $x \in X$, $y \in Y$.

$$(\exists x \in X,\ \forall y \in Y,\ P(x, y)) \quad \Longrightarrow \quad (\forall y \in Y,\ \exists x \in X,\ P(x, y)).$$

1. Obtain a witness x_0 that works for every y.
2. Fix an arbitrary y.
3. Use that same x_0; the universal hypothesis gives $P(x_0, y)$.

The converse can fail. Over $\mathbb{Z}$, take $P(x, y)$ to mean $x > y$. For each y, $x = y + 1$ works; no one integer is greater than every integer.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Example 5.1.6; converse counterexample is an extension.

Negating existence: two direct implications

$$\neg \exists x \in A,\ P(x) \quad \Longleftrightarrow \quad \forall x \in A,\ \neg P(x).$$

Forward

Assume no witness exists. For an arbitrary $a \in A$, if $P(a)$ held, a would be a witness. Thus $\neg P(a)$.

Backward

Assume every $a \in A$ fails P. If a witness w existed, we would have both $P(w)$ and $\neg P(w)$.

This equivalence also holds when A is empty. No choice of a default element is needed.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Example 5.1.6.

5.1 Practice: build both conversions

1. Prove $(P \wedge P) \leftrightarrow P$.
2. Prove $\neg(P \vee Q) \leftrightarrow (\neg P \wedge \neg Q)$.
3. Suppose $\forall x \in A,\ P(x) \leftrightarrow Q(x)$. Prove

$$(\exists x \in A,\ P(x)) \leftrightarrow (\exists x \in A,\ Q(x)).$$

In item 3, must the witness change when you convert the statement?

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Exercises 5.1.7.

5.1 Solution ideas

1. Forward: keep one copy of P. Backward: use the proof of P twice to build $P \wedge P$.
2. Forward: a proof of either P or Q would give the forbidden disjunction. Backward: split a hypothetical $P \vee Q$; each branch contradicts the corresponding negation.
3. Forward: obtain w and $P(w)$, apply the forward implication at w, and keep w as the witness for Q. Backward: use the reverse implication at the same witness.

The common structure is to unpack the given statement and rebuild the target.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.1, Exercises 5.1.7.

5.2: A case split need not decide the proposition

Law of the excluded middle

For every proposition P, we may reason by the two cases P and $\neg P$.

To prove a target Q:

1. Assume P and prove Q.
2. Assume $\neg P$ and prove Q.
3. Conclude Q, because the cases cover all possibilities.

In Lean, `by_cases h : P` creates these two branches. This is a classical logical principle, not an algorithm for deciding arbitrary propositions.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Examples 5.2.5 and 5.2.6.

A predicate with easy endpoints and a difficult middle

Following the textbook, call $k \in \mathbb{N}$ *superpowered* when

$$S(k): \quad \forall n \in \mathbb{N}, \quad k^{k^n} + 1 \text{ is prime.}$$

Exponentiation is grouped as $k^{(k^n)}$; the convention is $k^0 = 1$.

k	Useful calculation	Conclusion
0	$0^{0^0} + 1 = 1$	$\neg S(0)$
1	$1^{1^n} + 1 = 2$ for every n	$S(1)$
3	$3^{3^0} + 1 = 4$	$\neg S(3)$

For $k = 2$, the first values are $3, 5, 17, 257, 65537$. A finite list of successes does not prove $S(2)$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Examples 5.2.1–5.2.4.

Prove a boundary exists without locating it first

We know $S(1)$ and $\neg S(3)$. Prove

$$\exists k \in \mathbb{N}, \quad S(k) \wedge \neg S(k+1).$$

Case	Choose	Why it works
$S(2)$	$k = 2$	$S(2)$ and $\neg S(3)$
$\neg S(2)$	$k = 1$	$S(1)$ and $\neg S(2)$

Both cases produce a witness. This argument proves existence even before we determine which case holds.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Example 5.2.5.

The same boundary argument in abstract Lean

```
-- Demonstration: the argument uses only S 1 and Not (S 3).
example (S : Nat -> Prop) (h1 : S 1) (h3 : Not (S 3)) :
    Exists (fun k => And (S k) (Not (S (k + 1)))) := by
  by_cases h2 : S 2
  . use 2
    constructor
    . exact h2
    . exact h3
  . use 1
    constructor
    . exact h1
    . exact h2
```

The logical structure does not depend on the arithmetic inside S.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Example 5.2.5; abstract teaching adaptation.

Double negation and contraposition

Double negation elimination

Given $\neg\neg P$, split into P or $\neg P$. The first case proves P directly. The second contradicts $\neg\neg P$. Thus P follows classically.

Contraposition

$$(P \to Q) \quad \iff \quad (\neg Q \to \neg P).$$

Forward: assuming $\neg Q$ and P, the implication supplies the forbidden Q. Backward: assume P. If Q failed, the contrapositive would give $\neg P$; hence Q holds by classical reasoning.

The forward implication is direct; the reverse uses double negation elimination.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Example 5.2.6 and Exercises 5.2.7.

5.2 Practice: identify the classical step

1. From $\neg P \to \neg Q$ and Q, prove P. Show explicitly where excluded middle is used.
2. You are given the exact factorization

$$2^{2^5} + 1 = 4294967297 = 641 \cdot 6700417.$$

   Use it to select a concrete k satisfying $S(k) \wedge \neg S(k+1)$, where S is the superpowered predicate.

Why are five prime values insufficient to establish a universal statement?

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Exercises 5.2.7.

5.2 Solution ideas

1. Split into P and $\neg P$. In the first case, we are done. In the second, the given implication produces $\neg Q$, contradicting Q. Excluded middle licenses that case split.
2. The value at $n = 5$ has a nontrivial factor 641, so $\neg S(2)$. We already know $S(1)$; therefore choose $k = 1$.

What a counterexample must provide

To refute $\forall n,\ P(n)$, one failing input is enough. To prove it, checking a fixed finite sample is generally insufficient.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.2, Exercises 5.2.7.

5.3: Push negations through propositional operations

Original	Equivalent form	Textbook Lean lemma
$\neg\neg P$	P	`not_not`
$\neg(P \vee Q)$	$\neg P \wedge \neg Q$	`not_or`
$\neg(P \wedge Q)$	$\neg P \vee \neg Q$	`not_and_or`
$\neg(P \to Q)$	$P \wedge \neg Q$	`not_imp`

Interpret the last row

An implication fails when its premise holds and its conclusion fails. Negating an implication does *not* negate both sides.

These are classical equivalences; some directions require excluded middle.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Table 5.1.

Negating quantifiers changes who chooses the object

$$\neg \exists x \in A,\ P(x) \quad \Longleftrightarrow \quad \forall x \in A,\ \neg P(x),$$
$$\neg \forall x \in A,\ P(x) \quad \Longleftrightarrow \quad \exists x \in A,\ \neg P(x).$$

- No witness works $\Rightarrow$ every candidate fails.
- Not every candidate works $\Rightarrow$ there is a counterexample.

For ordered numbers, also use

$$\neg(a < b) \Longleftrightarrow b \leq a, \qquad \neg(a \leq b) \Longleftrightarrow b < a.$$

The domain of each quantifier stays the same throughout the transformation.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Table 5.1 and Example 5.3.3.

Normalize one connective at a time

Start with

$$\neg(\forall m \in \mathbb{Z},\ m \neq 2 \Rightarrow \exists n \in \mathbb{Z},\ n^2 = m).$$

1. Negate the universal quantifier:

$$\exists m \in \mathbb{Z},\ \neg(m \neq 2 \Rightarrow \exists n \in \mathbb{Z},\ n^2 = m).$$

2. Negate the implication:

$$\exists m \in \mathbb{Z},\ m \neq 2 \wedge \neg\exists n \in \mathbb{Z},\ n^2 = m.$$

3. Negate the existential quantifier:

$$\exists m \in \mathbb{Z},\ m \neq 2 \wedge \forall n \in \mathbb{Z},\ n^2 \neq m.$$

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Example 5.3.2.

Why negating “and” requires a case split

Prove $\neg(P \wedge Q) \Rightarrow \neg P \vee \neg Q$.

1. Suppose $\neg(P \wedge Q)$.
2. By excluded middle, either P or $\neg P$.
3. If $\neg P$, choose the left alternative.
4. If P, show $\neg Q$: an assumed Q would combine with P to give the forbidden $P \wedge Q$. Choose the right alternative.

Conversely, if either P or Q is false, they cannot both hold.

Lean connection

Use `constructor`, then `by_cases hP : P` in the forward direction; use `obtain` in the reverse direction.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Example 5.3.1.

A chained inequality hides a conjunction

The statement $n^2 < m < (n+1)^2$ means

$$n^2 < m \ \wedge \ m < (n+1)^2.$$

Therefore

$$\neg(\forall n \in \mathbb{Z},\ \exists m \in \mathbb{Z},\ n^2 < m < (n+1)^2)$$

is equivalent to

$$\exists n \in \mathbb{Z},\ \forall m \in \mathbb{Z}, \quad m \leq n^2 \ \vee \ (n+1)^2 \leq m.$$

To *prove* the resulting statement, choose $n = 0$. Every integer m satisfies $m \leq 0$ or $1 \leq m$.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Example 5.3.3; witness proof is an extension.

Normalize first, then supply the mathematical idea

```
-- Demonstration in the textbook's tactic environment.
example : Not (forall x : Real, x ^ 2 >= x) := by
  push_neg
  use (1 / 2 : Real)
  numbers
```

After `push_neg`, the goal asks for a real x with $x^2 < x$. The tactic exposes the witness obligation; the choice $x = \frac{1}{2}$ is the mathematical step.

What normalization does

It changes the form of the statement by equivalences. It does not automatically find the counterexample or prove the resulting inequality.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Example 5.3.4 and Exercises 5.3.6.

5.3 Practice: normalize, then prove or refute

1. Move all negations inward:

$$\neg\big(\forall a, b \in \mathbb{Z},\ ab = 1 \Rightarrow (a = 1 \vee b = 1)\big).$$

   Then decide whether the statement is true.

2. Normalize

$$\neg\big(\exists x \in \mathbb{R},\ \forall y \in \mathbb{R},\ y \leq x\big),$$

   and prove the normalized statement.

Keep the variable domains visible while changing the logical symbols.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Exercises 5.3.6.

5.3 Solution ideas

1. The normalized form is

$$\exists a, b \in \mathbb{Z}, \quad ab = 1 \wedge a \neq 1 \wedge b \neq 1.$$

   It is true: choose $a = b = -1$. Their product is 1, and neither equals 1.

2. The normalized form is

$$\forall x \in \mathbb{R},\ \exists y \in \mathbb{R}, \quad y > x.$$

   For an arbitrary x, choose $y = x + 1$. Then $y > x$.

A successful normalization turns the original wording into explicit inputs and witnesses.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Exercises 5.3.6.

One theorem, two useful proof presentations

$$\neg\exists n \in \mathbb{N},\ n^2 = 2 \quad \Longleftrightarrow \quad \forall n \in \mathbb{N},\ n^2 \neq 2.$$

Proof of the normalized form

Fix $n \in \mathbb{N}$. Either $n \leq 1$ or $n \geq 2$.

$$n \leq 1 \Rightarrow n^2 \leq 1 < 2, \qquad n \geq 2 \Rightarrow n^2 \geq 4 > 2.$$

Either strict inequality implies $n^2 \neq 2$.

The numerical argument is unchanged from Chapter 4, but normalization lets us begin with an arbitrary input rather than a hypothetical witness.

Source: Heather Macbeth, *The Mechanics of Proof*, Section 5.3, Example 5.3.5.

Review: four questions before manipulating a statement

1. What is the outermost connective or quantifier?
2. Am I proving an implication, an equivalence, or the truth of a particular formula?
3. Which variables may a witness depend on?
4. Does this direction use excluded middle?

A reliable workflow

Write the domains $\longrightarrow$ expose hidden conjunctions $\longrightarrow$ normalize one layer at a time $\longrightarrow$ identify the resulting proof obligations.

A truth table checks propositional structure; a quantified claim also needs reasoning about its objects.

Source: Heather Macbeth, *The Mechanics of Proof*, Chapter 5, Sections 5.1–5.3.

Sources and further work

Heather Macbeth, *The Mechanics of Proof*, Chapter 5.

- 5.1: logical equivalence, truth tables, and quantifiers.
- 5.2: excluded middle, the superpowered predicate, and contraposition.
- 5.3: Table 5.1 and worked negation normalizations.

Read Chapter 5 and practice the remaining logical equivalences.

Teaching prose, tables, and added counterexamples are original. Source comments record example locations. Lean snippets use the textbook's educational tactics.